

AVANCE DEL PROYECTO

ESTRUCTURA DE BASE DE DATOS



Integrantes:

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Objetivo:

El objetivo desarrollar un programa en Java que simule un sistema de gestión de tareas para una pequeña empresa. Este sistema utilizará diferentes estructuras de datos (pilas, colas y listas) para manejar tareas asignadas a distintos departamentos. (Practicar la implementación de estas estructuras y entender sus aplicaciones prácticas).

Diagramas de flujo Cliente:

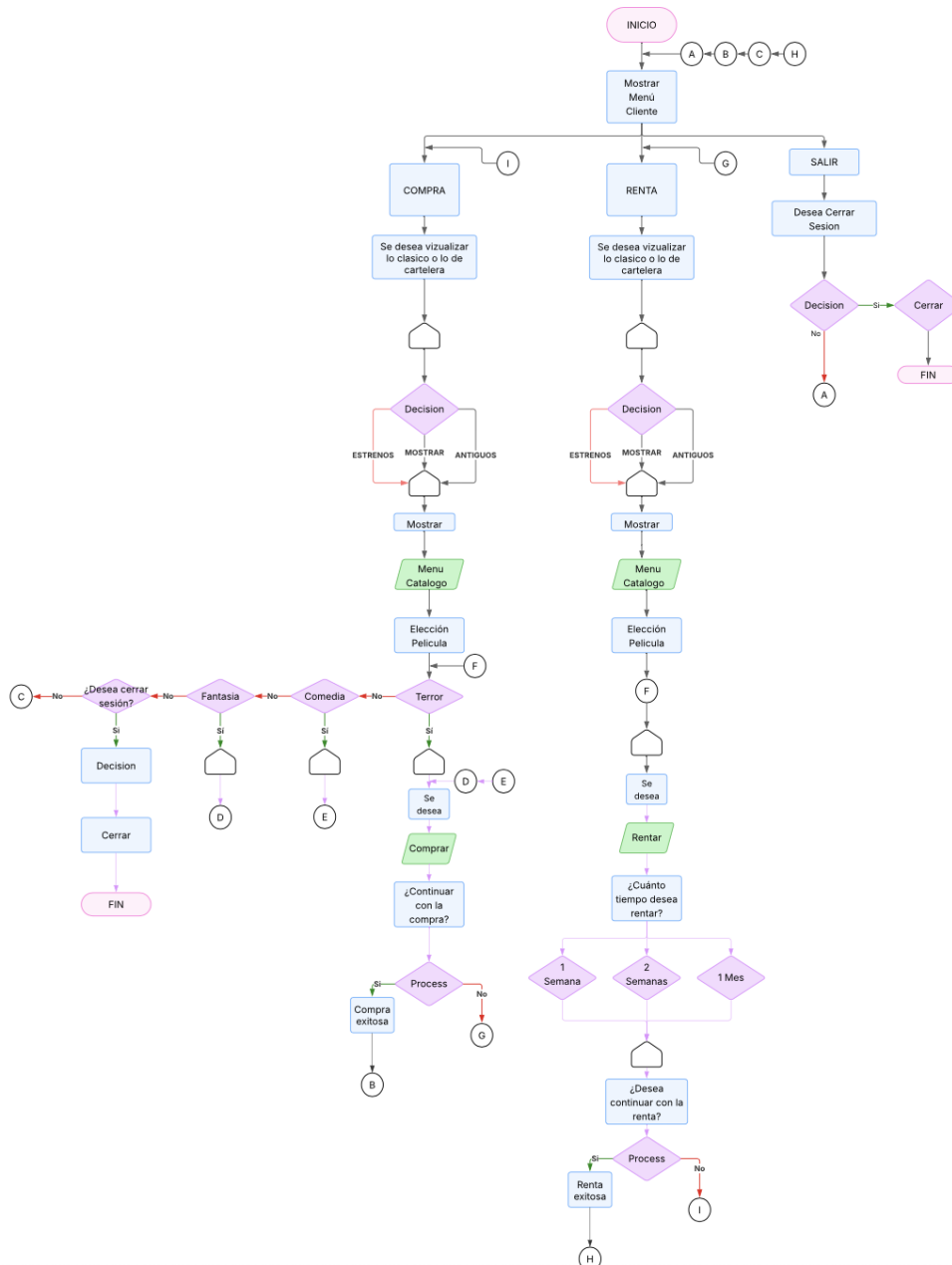


Diagrama de flujo Empleado:

Dentro de nuestro código podemos observar que se utilizaron múltiples datos tipo objeto (*Tasks*, *Worker*, *Client*, *Peliculas*), esto con la finalidad de poder agendarles a cada objeto las cualidades necesarias para el funcionamiento de este programa, cada objeto tiene métodos con los cuales podemos crear nuevos objetos del mismo tipo de una manera más accesible y rápida.

Para este programa implementamos un sistema de inicio sesión dependiendo del rol a desempeñar dentro del programa, tanto los clientes como los trabajadores pueden iniciar sesión dentro del programa o registrarse, de esta manera manteniendo un movimiento flexible y amistoso con los usuarios; el sistema de sesión nos permitió separar a los clientes de los trabajadores y que cada rol desempeñe sus debidas labores.

Código:

```
public class CLI_Client {
    private ClientManager clientManager;
    private Scanner scanner = new Scanner(System.in);
    private CLI cli = new CLI();

    public void menu(){
        System.out.println(x:"+-----+");
        System.out.println(x:"|          Bienvenido          |");
        System.out.println(x:"+-----+");
        System.out.println(x:"");
        System.out.println(x:"+-----+");
        System.out.println(x:"|      Seleccione una opción.      |");
        System.out.println(x:"| 1. Mostrar películas disponibles. |");
        System.out.println(x:"| 2. Agregar película al carrito.   |");
        System.out.println(x:"| 3. Eliminar película del carrito. |");
        System.out.println(x:"| 4. Mostrar carrito.               |");
        System.out.println(x:"| 5. Cerrar sesión.                |");
        System.out.println(x:"+-----+");
    }
}
```

```
public void start(){
    clientManager = new ClientManager();
    String option;
    do {
        menu();
        System.out.println(x:"");
        System.out.print(s:"Escriba el número aquí: ");
        option = scanner.nextLine();
        switch (option) {
            case "1":
                cli.menu2();
                break;
            case "2":
                addToCart();
                break;
            case "3":
                removeFromCart();
                break;
            case "4":
                showCart();
                break;
            case "5":
                System.out.println(x:"Gracias por visitarnos...");
                break;
            default:
                System.out.println(x:"Opción no válida, vuelva a intentarlo.");
                break;
        }
    }
}
```

```
public class CLI_Client {
    private void addToCart(){
        System.out.println(x:"");
        System.out.println(x:"+-----+");
        System.out.println(x:"|      Agregar al carrito      |");
        System.out.println(x:"+-----+");
        System.out.println(x:"");
        System.out.println(x:"+-----+");
        System.out.println(x:"|      Ingrese el titulo.      |");
        System.out.println(x:"+-----+");
        System.out.println(x:"");
        System.out.print(s:"Escribe aqui: ");
        String titulo = scanner.nextLine();
        clientManager.addToCart(titulo);
    }

    private void removeFromCart(){
        System.out.println(x:"");
        System.out.println(x:"+-----+");
        System.out.println(x:"|      Eliminar del carrito      |");
        System.out.println(x:"+-----+");
        System.out.println(x:"");
        System.out.println(x:"+-----+");
        System.out.println(x:"|      Ingrese el titulo.      |");
        System.out.println(x:"+-----+");
        System.out.println(x:"");
        System.out.print(s:"Escribe aqui: ");
        String titulo = scanner.nextLine();
        clientManager.removeFromCart(titulo);
    }
}
```

```
private void showCart(){
    for (Peliculas pelicula : clientManager.viewCart()){
        System.out.println(x:"");
        System.out.println(x:"+-----+");
        System.out.println(x:"| Titulo: " + pelicula.getTitulo() + " |");
        System.out.println(x:"| Genero: " + pelicula.getGenero() + " |");
        System.out.println(x:"| Año: " + pelicula.getAño() + " |");
        System.out.println(x:"+-----+");
    }
}
```

```
public class CLI_Login {
    private Scanner scanner = new Scanner(System.in);
    private CLI_Client cliClient = new CLI_Client();
    private CLI_Worker cliWorker = new CLI_Worker();
    private CLI cli = new CLI();
    private WorkerManager workerManager = new WorkerManager();
    private ClientManager clientManager = new ClientManager();
}
```

```
public void start(){
    String command;
    do{
        System.out.println(x:"");
        menu();
        System.out.print(s:"Escriba la opción aquí: ");
        command = scanner.nextLine();
        switch (command) {
            case "1":
                workerLogin();
                break;
            case "2":
                clientLogin();
                break;
            case "3":
                cli.start();
                break;
            case "4":
                System.out.println(x:"Gracias por visitarnos.");
                break;
            default:
                System.out.println(x:"Opción no valida.");
                break;
        }
    }while(!command.equals(anObject:"4"));
}
```

```
private void menu(){
    System.out.println(x:"+-----+");
    System.out.println(x:"|          Bienvenido.          |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|      Seleccione su rol.      |");
    System.out.println(x:"|  1. Trabajador.              |");
    System.out.println(x:"|  2. Cliente.                 |");
    System.out.println(x:"|  3. Administrador.           |");
    System.out.println(x:"|  4. Salir.                    |");
    System.out.println(x:"+-----+");
}
```

```
private void workerLogin(){
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|      Iniciar Sesión.      |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|      Ingrese sus datos.      |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    System.out.print(s:"Nombre: ");
    String name = scanner.nextLine();
    System.out.println(x:"");
    System.out.println(x:"Departamento: ");
    System.out.println(x:" 1. Finanzas.");
    System.out.println(x:" 2. Inventario.");
    System.out.print(s:"-> ");
    int department = scanner.nextInt();
    scanner.nextLine();
    System.out.println(x:"");
    System.out.print(s:"ID: ");
    int id = scanner.nextInt();
    scanner.nextLine();
}
```

```
if (workerManager.login(name, department, id) == false){
    System.out.println(x:"Parece que no estas registrado, se te registrará de inmediato.");
    workerManager.addWorker(name, department, id);
    System.out.println(x:"");
    System.out.println(x:"Has sido registrado con exito.");
    cliWorker.start();
}
cliWorker.start();
```

```
private void clientLogin(){
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|      Iniciar Sesión.      |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|      Ingrese sus datos.      |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    System.out.print(s:"Nombre: ");
    String name = scanner.nextLine();
    System.out.println(x:"");
    System.out.print(s:"Contraseña: ");
    String password = scanner.nextLine();
    if (clientManager.login(name, password) == false){
        System.out.println(x:"Parece que eres nuevo cliente, permitenos registrarte de inmediato.");
        clientManager.addClient(name, password);
        System.out.println(x:"");
        System.out.println(x:"Listo, has sido registrado con exito.");
        cliClient.start();
    }
    cliClient.start();
}
```



```
public class WorkerManager{
    HashMap<Integer, Task> tasksFinance = new HashMap<>();
    HashMap<Integer, Task> tasksInventory = new HashMap<>();
    HashMap<Integer, Worker> workers = new HashMap<>();

    public WorkerManager(){
        workers.put(key:0104, new Worker(name:"Victor", department:1, id:0104));
        workers.put(key:2804, new Worker(name:"Sebastian", department:1, id:2804));
        workers.put(key:1411, new Worker(name:"Michel", department:2, id:1411));
        workers.put(key:2311, new Worker(name:"Daniela", department:2, id:2311));
    }

    public void addTask(String description, int department, int id, int priority){
        if (department == 1) {
            tasksFinance.put(id, new Task(id, description, priority));
        }else if (department == 2) {
            tasksInventory.put(id, new Task(id, description, priority));
        }
    }

    public boolean login(String name, int department, int id){
        for (Worker worker : workers.values()){
            if (worker.getName().equals(name) && worker.getDepartment() == department && worker.getId() == id){
                return true;
            }
        }
        return false;
    }
}
```

```
public void addWorker(String name, int department, int id){
    if(!workers.containsKey(id)){
        workers.put(id, new Worker(name, department, id));
    }else {
        System.out.println("El trabajador con ID: " + id + " ya existe.");
    }
}

public void removeWorker(int id){
    if(workers.containsKey(id)){
        workers.remove(id);
        System.out.println("Trabajador: " + workers.get(id).getId() + " - " + workers.get(id).getName() + " - " + workers.get(id).getDe
    }
}
```

```
public void removeTask(int department, int id){
    if (department == 1) {
        if (tasksFinance.containsKey(id)) {
            tasksFinance.remove(id);
        }else {
            System.out.println(x:"Tarea no encontrada.");
        }
    }else if (department == 2) {
        if (tasksInventory.containsKey(id)) {
            tasksInventory.remove(id);
        }else {
            System.out.println(x:"Tarea no encontrada.");
        }
    }else {
        System.out.println(x:"Departamento no encontrado.");
    }
}
```



```
public List<Task> viewTasks(int department, int priority){
    List<Task> tasks = new ArrayList<>();
    if (department == 1) {
        for (Task task : tasksFinance.values()) {
            if (task.getPriority() == priority) {
                tasks.add(task);
            }
        }
    }else if (department == 2) {
        for (Task task : tasksInventory.values()) {
            if (task.getPriority() == priority) {
                tasks.add(task);
            }
        }
    }else {
        System.out.println(x:"Departamento no encontrado.");
    }
    return tasks;
}
```

```
public ClientManager(){
    clients.add(new Client(name:"Victor", password:"1234"));
    clients.add(new Client(name:"Sebastian", password:"1234"));
    clients.add(new Client(name:"Daniela", password:"1234"));
    clients.add(new Client(name:"Michel", password:"1234"));
}

public boolean login(String name, String password){
    for (Client client : clients){
        if (client.getName().equals(name) && client.getPassword().equals(password)){
            return true;
        }
    }
    return false;
}

public void addToCart(String titulo){
    for (Películas pelicula : películas.values()){
        if (pelicula.getTitulo().equalsIgnoreCase(titulo)){
            cart.add(pelicula);
            System.out.println("Pelicula agregada al carrito: " + pelicula.getTitulo());
        }
    }
    System.out.println("Pelicula no encontrada: " + titulo);
}
```

```
public List<Películas> viewCart(){
    return cart;
}

public void removeFromCart(String titulo){
    for(Películas pelicula : cart){
        if(pelicula.getTitulo().equalsIgnoreCase(titulo)){
            cart.remove(pelicula);
            System.out.println("Película: " + pelicula.getTitulo() + " eliminada del carrito.");
            break;
        }
    }
    System.out.println("Película no encontrada: " + titulo);
}

public void addClient(String name, String password){
    clients.add(new Client(name, password));
}
```

```
public class CLI_Worker {
    private WorkerManager workerManager = new WorkerManager();
    private Scanner scanner = new Scanner(System.in);

    public void start(){
        String command;
        do {
            System.out.println(x:"");
            menu();
            System.out.println(x:"");
            System.out.print(s:"Ingrese su opción: ");
            command = scanner.nextLine();
            switch (command) {
                case "1":
                    addTask();
                    break;
                case "2":
                    removeTask();
                    break;
                case "3":
                    showTask();
                    break;
                case "4":
                    System.out.println(x:"Gracias por su labor del día.");
                    break;
                default:
                    System.out.println(x:"Opción no válida, intente de nuevo.");
                    break;
            }
        } while (true);
    }
}
```

```
public void menu(){
    System.out.println(x:"+-----+");
    System.out.println(x:"| Bienvenido |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"| Seleccione una opción. |");
    System.out.println(x:"| 1. Añadir tarea. |");
    System.out.println(x:"| 2. Remover tarea. |");
    System.out.println(x:"| 3. Mostrar tareas segun prioridad. |");
    System.out.println(x:"| 4. Cerrar sesión. |");
    System.out.println(x:"+-----+");
}
```

```
private void addTask(){
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|      Añadir tarea.      |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|      Ingrese la tarea.   |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    String description = scanner.nextLine();
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|      Ingrese su departamento. |");
    System.out.println(x:"| 1. Finanzas.              |");
    System.out.println(x:"| 2. Inventario.           |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    int department = scanner.nextInt();
    scanner.nextLine();
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|      Ingrese el ID de la tarea. |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    int id = scanner.nextInt();
    scanner.nextLine();
```

```
System.out.println(x:"");
System.out.println(x:"+-----+");
System.out.println(x:"|      Ingrese el nivel de prioridad. |");
System.out.println(x:"| 1. Alta.                          |");
System.out.println(x:"| 2. Media.                         |");
System.out.println(x:"| 3. Baja.                          |");
System.out.println(x:"+-----+");
System.out.println(x:"");
int priority = scanner.nextInt();
scanner.nextLine();
workerManager.addTask(description, department, id, priority);
```

```
private void removeTask(){
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|          Remover tarea.          |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|    Ingrese su departamento.    |");
    System.out.println(x:"| 1. Finanzas.                    |");
    System.out.println(x:"| 2. Inventario.                  |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    int department = scanner.nextInt();
    scanner.nextLine();
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|    Ingrese el ID de la tarea.    |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    int id = scanner.nextInt();
    scanner.nextLine();
    workerManager.removeTask(department, id);
}
```

```
private void showTask(){
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|          Mostrar tareas.          |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|    Ingrese su departamento.    |");
    System.out.println(x:"| 1. Finanzas.                    |");
    System.out.println(x:"| 2. Inventario.                  |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    int department = scanner.nextInt();
    scanner.nextLine();
    System.out.println(x:"");
    System.out.println(x:"+-----+");
    System.out.println(x:"|    Ingrese el nivel de prioridad. |");
    System.out.println(x:"| 1. Alta.                        |");
    System.out.println(x:"| 2. Media.                       |");
    System.out.println(x:"| 3. Baja.                        |");
    System.out.println(x:"+-----+");
    System.out.println(x:"");
    int priority = scanner.nextInt();
    scanner.nextLine();
}
```

```
int priority = scanner.nextInt();
scanner.nextLine();
for (Task task : workerManager.viewTasks(department, priority)){
    System.out.println(x:"+-----+");
    System.out.println(x:"| ID: " + task.getId() + "          |");
    System.out.println(x:"| Descripción: " + task.getDescription() + " |");
    System.out.println(x:"| Prioridad: " + task.getPriority() + "      |");
    System.out.println(x:"+-----+");
}
```

```
public Client(String name, String password){
    setName(name);
    setPassword(password);
}

public String getName(){
    return name;
}

public String getPassword(){
    return password;
}

public void setName(String name){
    if (name == null || name.trim().isEmpty()){
        throw new IllegalArgumentException(s:"El nombre no puede estar vacio.");
    }
    this.name = name;
}
```

```
public void setPassword(String password){
    if (password == null || password.trim().isEmpty()){
        throw new IllegalArgumentException(s:"La contraseña no puede estar vacia.");
    }
    this.password = password;
}
```

```
public Task(int id, String description, int priority){
    setId(id);
    setDescription(description);
    setPriority(priority);
}

public int getId(){
    return id;
}

public String getDescription(){
    return description;
}

public int getPriority(){
    return priority;
}
```

```
public void setId(int id){
    if(id <= 0){
        throw new IllegalArgumentException(s:"ID invalido.");
    }
    this.id = id;
}

public void setDescription(String description){
    if(description == null || description.trim().isEmpty()){
        throw new IllegalArgumentException(s:"La descripcion no puede estar vacia.");
    }
    this.description = description;
}

public void setPriority(int priority){
    if(priority < 1 || priority > 3){
        throw new IllegalArgumentException(s:"Nivel de prioridad invalido.");
    }
    this.priority = priority;
}
```

```
public Worker(String name, int department, int id){
    setName(name);
    setDepartment(department);
    setID(id);
}

public String getName(){
    return name;
}

public int getDepartment(){
    return department;
}

public int getId(){
    return id;
}
```

```
public void setName(String name){
    if(name == null || name.trim().isEmpty()){
        throw new IllegalArgumentException(s:"El nombre no puede estar vacio.");
    }
    this.name = name;
}

public void setDepartment(int department){
    if(department > 2 || department < 1){
        throw new IllegalArgumentException(s:"El departamento seleccionado no es valido.");
    }
    this.department = department;
}

public void setID(int id){
    if(id <= 0){
        throw new IllegalArgumentException(s:"El ID debe ser un numero positivo.");
    }
    this.id = id;
}
```

Captura Menú:

```
+-----+
|           Bienvenido.           |
+-----+

+-----+
|  Seleccione su rol.             |
| 1. Trabajador.                 |
| 2. Cliente.                   |
| 3. Administrador.              |
| 4. Salir.                      |
+-----+
```

```
+-----+
|           Bienvenido           |
+-----+

+-----+
|  Seleccione una opción.         |
| 1. Añadir tarea.               |
| 2. Remover tarea.              |
| 3. Mostrar tareas segun prioridad. |
| 4. Cerrar sesión.             |
+-----+
```



```
+-----+
|           Bienvenido           |
+-----+

+-----+
| Seleccione una opción.         |
| 1. Mostrar peliculas disponibles. |
| 2. Agregar pelicula al carrito.   |
| 3. Eliminar pelicula del carrito. |
| 4. Mostrar carrito.              |
| 5. Cerrar sesión.                |
+-----+
```